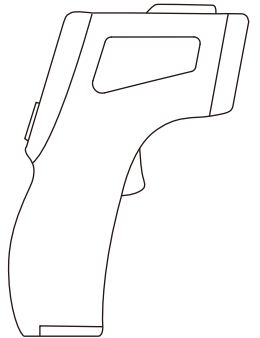


Non-contact Infrared Forehead Thermometer

Product Manual



Epidemic Situation Emergency Products

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1. Description

This product is a non-contact infrared forehead thermometer that professionally measures the temperature of the forehead of the human body. According to the difference of human skin, the measured temperature will be different; the temperature caused by different parts of the human body will be different because the more exposed the human body The part is more affected by the ambient temperature.

2. Use safety manual

- Please read this manual carefully before use.
- The operating temperature of this product is 16 ℃ ~ 35 ℃, the optimal temperature is 25 ℃.
- Do not use this product in an environment above 50 ° C or below 0 ° C.
- Do not place this product near charged objects to avoid electric shock.
- Please do not use this product in an environment with relative humidity greater than 85%.
- Do not place this product too close to the electromagnetic range. (E.g. radio, mobile phone, etc.).
- Please do not expose this product to the sun, or near the stove, or contact it with water.
- Do not bump or drop the product, and do not use it if it is damaged.
- Sweat, hair, hat or scarf on forehead can affect the accuracy of the measurement.
- Make sure the measurement distance is within 5 cm.
- When forehead sweating or other reasons cause the forehead temperature does not reflect the body temperature normally, please measure from the earlobe.
- When cleaning is required, wipe the surface of the meter lightly with alcohol.
- Contact the distributor if there is any problem with the product and cannot repair the product by yourself.
- It is forbidden to take body temperature measurement when the ambient temperature changes greatly.

3. Features

- 1) Non-contact high-precision temperature measurement
- 2) Choice of ℃ or °F

Setting method: Press this button"for 8 seconds to switch.

- 3) Alarm value can be set (the preset value of this product is 38 ℃)

Setting method: set by default

- 4) Buzzer function (can be on or off)

Setting method: Press this button"once to turn on the buzzer.

- 5) LCD display with backlight can be used by users in dark

Setting method: Press this button"once to turn on the backlight.

Press this button"for 8 seconds to convert to object measurement, LCD display"temp"; and then press again for 8 seconds to switch to the body measurement, LCD display"Body temp".

- 6) Automatic range selection; resolution is 0.1℃ (0.1°F)
- 7) Can store the latest 32 measurement data (press the up and down arrow keys to view the latest 32 measurement data that has been stored)
- 8) Automatic data retention and automatic shutdown.

4. Other instructions

Non-contact infrared forehead thermometer is a professional thermometer for measuring human forehead. It is widely used in home use. This product cannot replace the diagnosis of doctors.

Measurement completion prompt function:

Full display at power-on, backlight on, "----" displayed, ℃ flashing;

Switch on: "Bi" sound ; Enter the measurement: "Bi" sound;

The temperature measurement is completed, the temperature is lower than the fever alarm point,

"Bi" sound, and the temperature is within the range of 38℃~ 42.9℃, sounds six times.

Fever prompt function:

In the body temperature mode, the electronic thermometer should have an over-temperature reminder

function. When the measurement temperature is greater than or equal to 38 ° C, there should be an audible prompt, and there is a"Bi Bi Bi Bi Bi Bi" sound.

5. Precautions Before Use

Power On Self Test

△Aim at the measured object, press the measurement switch, the LCD will display all the numbers and characters of the self-test, and the self-test screen appears. This screen is displayed for about 1 second.

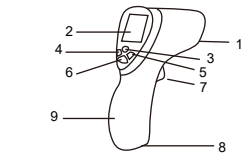
△After the power-on self-test has been completed, an

6. Product Structure

Model: GQ-129

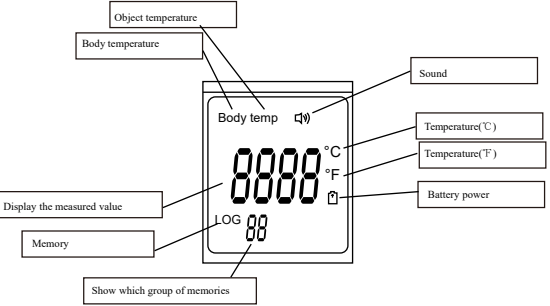
Structural composition: It is mainly composed of infrared probe assembly, main circuit board assembly, LCD display assembly and housing assembly.

- ①. Infrared sensor
- ②. LCD display
- ③. Backlight on / off button
- ④. Up button (view historical data)
- ⑤. Down button (view historical data)
- ⑥. Sound switch button
- ⑦. Measurement switch



- ⑧. Battery cover
- ⑨. Handle

7. LCD display description



8. Product scope and contraindications

8.1 The scope of products

The body temperature of the subject is displayed by measuring the heat radiation from the forehead.

8.2 Product contraindications

- 1) Conditions such as birth defects, congenital malformations, septic shock, and circulatory

- failure may severely affect forehead temperature measurement
- 2) Suffering from mental disorders.
 - 3) Suffering from severe heart, liver and kidney diseases.

9. Technical index

9.1 Basic parameters

Show exact bits	0.1℃ (0.1°F)
Power	DC 3V (2 AAA batteries in series)
Size	88 *46 *174mm
Weight	130g
Production date	See product nameplate for details

9.2 Measuring range

Human body mode temperature range	32.0~42.9℃
Measuring distance range	3-5cm (The best distance is 3 cm)
Automatic shut-down	About 30s

9.3 Measurement accuracy

32℃~34.9℃	±0.3℃
35℃~42℃	±0.2℃
42.1℃~42.9℃	±0.3℃

9.4 Service life

Product life is 2 years.

9.5 Software release version

V1.0

9.6 Work, storage, transportation, and environmental requirements

1) Operating environment:

- Operating temperature: 16℃~35℃, best 25℃;
- Relative humidity: ≤ 85%;
- Atmospheric pressure: 70kPa~106kPa
- Power: DC 3V (2 AAA batteries in series)

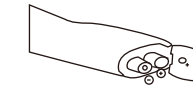
2) Storage and transportation environment:

- Storage temperature: 0℃~~+40℃;
- Relative humidity: 10%~85%;
- Non-corrosive gas, well-ventilated room.
- The transportation requirements are stipulated in the order contract, and the severe impact, vibration and rain and snow splashing during transportation must be prevented.

10. Operating instructions

10.1 Battery instructions

At the bottom of the handle, there is an oval with an arrow on it. Please push the battery cover forward with your hand in the direction of the arrow.



■ Note for battery installation and replacement:

- 1) After the battery cover is opened, place the battery in the direction of the positive and negative electrodes at the place where the batteries are installed. Pay attention to the correct orientation of the positive and negative electrodes.
- 2) Wait for 10 minutes to warm up for the first time or immediately after inserting a new battery.

- 3) When the battery power is low, the symbol will appear on the display, prompting to replace the new battery, open the battery compartment cover (see 6 product structure), pay attention to the positive and negative polarity when replacing the new battery. Improper placement may cause damage to the product.
- 4) Remove the battery when it has not been used for a long time to prolong its service life and prevent damage due to battery leakage.

10.2 Temperature measurement steps

1. Point the thermometer at the center of the forehead (above the center of the eyebrow) and keep it vertical, about 3-5 cm away. Press the measurement switch and the temperature will be displayed immediately.



2. When the measurement switch is pressed (see 6 Product Structure), the current measurement temperature value is displayed on the display, and the measurement data can be automatically stored (press the up and down arrow keys to check the latest 32 measured data that has been stored).

Note:

- 1) Before measuring, make sure that there is no hair, sweat, makeup or hats on it.
- 2) When forehead sweating or other reasons cause the forehead temperature does not reflect the body temperature normally, please measure after aiming at the earlobe, and make sure that there is no hair, sweat, cosmetics or hat covering.



- 3) The non-contact infrared forehead thermometer has not been used for a long time, the first time the machine is

opened for environmental temperature detection, and the boot time is extended by 1 to 2 seconds.

4) A person's body temperature changes at different times of the day and is also affected by other external conditions, such as age, gender, and skin color.

11. Reference temperature

11.1 Normal body temperature range at different measurement locations

The human body is a very complex biological comprehensive system. Body temperature is an important data indicating whether human life activities are normal. Usually we measure our health by measuring the temperature of the forehead, cochlea, anus, mouth and armpit. It is measured in different parts. Body temperature will vary, please refer to the table below for specific differences:

Measurement Parts	Normal temperature (°C)	Normal Fahrenheit (°F)
anus	36.6~38	97.8~100.4
Oral cavity	35.5~37.5	95.9~99.5
armpit	34.7~37.3	94.4~99.1
ear	35.8~38	96.4~100.4
forehead	36~37.2	97.4~98.4

11.2 Normal body temperature range at different ages

A person's body temperature changes at different times of the day, and is also affected by other external conditions, such as age, gender, skin color, fitness, etc. For the normal temperature range of different age groups, please refer to the table below:

Age	Normal temperature (°C)	Normal Fahrenheit (°F)
0~2 years	36.4~38.0	97.5~100.4
3~10 years	36.1~37.8	97.0~100.0
11~65 years	35.9~37.6	96.6~99.7
>65 years	35.8~37.5	96.4~99.5

Note:

Women's body temperature is different from men's, generally about 0.3 °C higher than men's, body

temperature during ovulation will rise 0.3 °C -0.5 °C than usual.

12. Advice

·The protective glass outside the LCD frame is very important, and it is also the fragile part of the instrument. Please use it carefully.

·Do not charge non-rechargeable batteries and do not throw them into fire.

·Do not expose the product to the sun, and do not touch the water.

13. Care and Maintenance

13.1 Product care and cleaning

- 1) The probe part is the most precise part of the product and must be carefully protected.
- 2) Do not use abrasive cleaners to clean the product.
- 3) Never immerse the product in water or other liquids.
- 4) Keep this product in a dry place to avoid dust, pollution and direct sunlight.
- 5) Please clean the non-contact electronic thermometer regularly, usually once a month. If necessary, you can clean it properly according to the actual soiling of the product. Use a dry soft cloth to clean the non-contact electronic thermometer. If the instrument is extremely dirty, use a cotton cloth or cotton swab with an alcohol content of 70% to clean and disinfect the case and sensor head (the infrared detector cannot be wiped). Be careful not to allow liquid to penetrate the inside of the thermometer.

necessary, you can clean it properly according to the actual soiling of the product. Use a dry soft cloth to clean the non-contact electronic thermometer. If the instrument is extremely dirty, use a cotton cloth or cotton swab with an alcohol content of 70% to clean and disinfect the case and sensor head (the infrared detector cannot be wiped). Be careful not to allow liquid to penetrate the inside of the thermometer.

13.2 Product Maintenance

If you encounter the following problems during use, please follow the instructions in the maintenance instructions to find a solution. If the problem persists, please contact our customer service.

- 1) LCD cannot display the value

If it is lower than 32 °C or higher than 42.9 °C in the body measurement state, the LCD will not display data, and display "Lo" or "Hi".

- 2) LCD display message "Hi"

Use non-contact electronic body temperature timer, LCD display message "Hi", indicating that it

has exceeded the measurement range or the temperature is higher than 42.9 °C in human temperature measurement mode.

- 3) LCD display message "Lo"

Using a non-contact electronic body temperature timer, the LCD displays the message "Lo", indicating that the measurement temperature is lower than the measurement range or the temperature

is lower than 32 °C in the human body temperature measurement mode.

The information "Lo" or "Hi" appears in the following situations for reference:

Reasons for "Lo" or "Hi"	Advice
Temperature value is affected by hair and sweat.	Ensure no obstructions during temperature measurement.
Temperature is affected by changes in airflow.	Ensure that the air remains stable during temperature measurements.
The measurement distance is too far.	Please note that the measurement distance should not be greater than 5 cm.
Enter indoors from low or high temperature outdoors.	Please wait for 20 minutes and wait for the subject's temperature to adapt to the measurement environment before measuring.

14. Waste disposal instructions

■ Disposing electronic products and batteries directly in the trash can cause harm to the environment. Please dispose of them in accordance with the laws in your area.

■ Do not discard the non-contact electronic thermometer at the end of its use. Please dispose of it according to the laws in your area, or contact the manufacturer for recycling.

15. Electromagnetic compatibility warning



Note:

- The non-contact electronic thermometer meets the requirements of YY0505-2012 standard electromagnetic compatibility.
- The user should install and use the electromagnetic compatibility information provided in the

random files.

- Portable and mobile RF communication equipment may affect the performance of non-contact electronic thermometers, and avoid strong electromagnetic interference when using, such as near mobile phones, microwave ovens, etc.
- Guidelines and manufacturer's declarations are detailed in the attachment.
- Warning:
- The equipment or system should not be used close to or stacked with other equipment. If it must be used close to or stacked, it should be observed to verify that it can operate normally in the configuration in which it is used.

15. Symbol Description

Symbol Graphics	Meaning	Symbol Graphics	Meaning
	Note, refer to the attached file		Refer to manual
	Low voltage alert		BF type application part
	Trash can		

16. Product accessories list

1 thermometer, 1 manual and 1 certificate.

EMC Declaration

Guidance and manufacturer's declaration – electromagnetic immunity			
The "GQ-129" is intended for use in the electromagnetic environment specified below. The customer or the user of the "GQ-129" should ensure that it is used in such an environment.			
Immunity test	IEC 60601 test level	Compliance level	Electromagnetic environment – guidance
Electrostatic discharge (ESD) IEC 61000-4-2	±6 kV contact ±8 kV air	±6 kV contact ±8 kV air	Floors should be wood, concrete or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30 %.

Electrical fast transient/ burst IEC 61000-4-4	±2 kV for power supply lines ±1 kV for input/	Not Applicable	Mains power quality should be that of a typical commercial or hospital environment.
Surge IEC 61000-4-5	±1 kV line(s) to line(s) ±2 kV line(s) to earth	Not Applicable	Mains power quality should be that of a typical commercial or hospital environment.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	<5 % UT (>95 % dip in UT) for 0.5 cycle 40 % UT (60 % dip in UT) for 5 cycles 70 % UT (30 % dip in UT) for 25 cycles <5 % UT (>95 % dip in UT) for 5 sec	Not Applicable	Mains power quality should be that of a typical commercial or hospital environment. If the user of the "GQ-129" requires continued operation during power mains interruptions, it is recommended that the "GQ-129" be powered from an uninterruptible power supply or a battery.
Power frequency (50/60 Hz) magnetic field IEC 61000-4-8	3 A/m	3 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.
NOTE UT is the a.c. mains voltage prior to application of the test level.			

Guidance and manufacturer's declaration – electromagnetic immunity			
The "GQ-129" is intended for use in the electromagnetic environment specified below. The customer or the user of the "GQ-129" should ensure that it is used in such an environment.			
Immunity test	IEC 60601 test level	Compliance level	Electromagnetic environment – guidance
			Portable and mobile RF communications equipment should be used no closer to any part of the "GQ-129", including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter. Recommended separation distance

Conducted RF IEC 61000-4-6	3 Vrms 150 kHz to 80 MHz	Not Applicable	$d = \left[\frac{3.5}{P} \right] \sqrt{P}$ V1 $d = \left[\frac{3.5}{P} \right] \sqrt{P}$ E1 $d = \left[\frac{3}{P} \right] \sqrt{P}$ E1 80MHz to 800MHz 800MHz to 2.5 GHz
Radiated RF IEC 61000-4-3	3 V/m 80 MHz to 2.5 GHz	3 V/m	where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in metres (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey, a should be less than the compliance level in each frequency range.b Interference may occur in the vicinity of equipment marked with the following symbol:

NOTE 1 At 80 MHz and 800 MHz, the higher frequency range applies. NOTE 2 These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.	
a	Field strengths from fixed transmitters, such as base stations for radio (cellular/ cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the "GQ-129" is used exceeds the applicable RF compliance level above, the Medical GQ-128,GQ-129 should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as reorienting or relocating the "GQ-129".
b	Over the frequency range 150 kHz to 80 MHz, field strengths should be less than 3 V/m.

Recommended separation distances between portable and mobile RF communications equipment and the Medical GQ-128,GQ-129			
The "GQ-129" is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the Medical GQ-128,GQ-129 can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the "GQ-129" as recommended below, according to the maximum output power of the communications equipment.			
Rated maximum output power of transmitter W	Separation distance according to frequency of transmitter m		
	150 kHz to 80 MHz	80 MHz to 800 MHz	800 MHz to 2.5 GHz
	0,01	/	0.12
	0,1	/	0.38
	1	/	2.3
10	/	3.8	7.3
100	/	12	23
For transmitters rated at a maximum output power not listed above, the recommended separation distance d in meters (m) can be estimated using the equation applicable to the frequency of the transmitter, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer. NOTE 1 At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies. NOTE 2 These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.			

Product technical requirement number:
Use period: 2 years

Warranty matters:

- 1.From the date of purchase, medical products can enjoy a one-year free warranty and lifetime maintenance with a shopping invoice;
- 2.We will not provide free warranty service for the following failures caused by user reasons. Such as:
A)Failure caused by unauthorized disassembly or modification.
B)Failure caused by accidental drop during use or handling.
C)Failure caused by lack of reasonable maintenance.
D)Failure caused by not following the correct instructions in the instruction manual.

- 3.When requesting a free warranty service, you must have a warranty card with the date of purchase and stamp of purchase (including store name and address). Please ask the staff to stamp the warranty card when buying this product;

- 4.Repair services outside the warranty will be charged as required;

- 5.When requesting warranty service, please take this product to the sales shop for repair.

Warranty Card

Product model: _____ Customer name _____

Purchase date: _____ Store Address: _____

Fault content: _____ Postal code: _____

Fault content: _____

Customer Address: _____

Repairman signed: _____